AMARNATH S. PATEL

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Empirical LLM evaluation, adversarial robustness, and applied AI systems — with the GPU and numerics background under it.

EDUCATION

University of Central Florida **Undergraduate Student 4.00 GPA**
B.S. Physics (Optics & Lasers) and Mathematics, Computer Science minor August 2025 – Expected May 2029

- **Machine learning and computing:** Deep Learning, Data Structures & Algorithms, Computer Architecture, Object-Oriented Programming, Discrete Structures
- **Mathematics:** Linear Algebra (proof-based, in progress), Complex Analysis, Partial Differential Equations (in progress), Applied Linear Algebra, Ordinary Differential Equations, Mathematical Methods for Physics

Florida Atlantic University **University coursework via FAU High School, ages 14–18 3.66 GPA**
111 credit hours completed between ages 14 and 18, alongside high school August 2021 – May 2025

RESEARCH EXPERIENCE

Undergraduate Researcher — FAU Grant-Funded AI Safety Research Project January 2024 – March 2025
Supervisor: Tucker Hindle, Florida Atlantic University

- Evaluated whether AI-text detection actually works, and what it costs to defeat it. Benchmarked 10 statistical coherence and detection measures — GPT-2 perplexity, RoBERTa classification, BERT next-sentence prediction, latent semantic analysis, NLI entailment, burstiness — on a 255,000-passage corpus of human and model-generated text.
- Established GPT-2 perplexity as the strongest discriminator (3.35× separation, 17.5 versus 58.5), RoBERTa as the only effective trained detector, and BERT NSP as carrying no usable signal for the task — a negative result worth having.
- Built four adversarial evasion pipelines against those detectors: iterative paraphrase-and-recheck, coherence- and perplexity-based tree-search decoding, list-branching, and a reinforcement-learning approach — over 3,125 generated sequences from Gemini 2.0 Flash, Gemini 1.5 Pro and GPT-4o. Characterized a consistent quality–evasion tradeoff: every method that reliably beat the detector degraded the text.
- Grant-funded with HPC access; technical report presented at the Wilkes Honors College Symposium, 2025.

Undergraduate Research Assistant — UCF Physics Department March 2026 – Present
Physics Education Research — Dr. Zhongzhou Chen (NSF Award 2421299; Gates Foundation INV-076932)

- Built an agentic generation–evaluation pipeline that drafts candidate physics problems and validates them against instructor criteria before human review, running at scale over the Azure OpenAI batch API. Presented the batch-processing approach to automated grading at UCF, June 2026.
- Built ESTELA (Rust, Tauri 2), the assessment tool the pipeline feeds: multi-version isomorphic exams from a structured bank of 615 problems across 29 banks, 13 topic areas and 11 question types, with export to four formats. In use by external adopters outside the university; manuscript in preparation.

Undergraduate Researcher — UCF Astrophotonics Lab, CREOL August 2025 – Present
Supervisor: Dr. Stephen Eikenberry

- Inference on noisy experimental data: FFT demodulation, filtering, and joint nonlinear optimization over three coupled parameters to recover complex amplitude and phase from raw interferograms, validated at ≈ 98% fidelity against simulated ground truth (NumPy/SciPy).
- Built the acquisition system that produces the data — four instruments driven as one experiment, with in-loop objective optimization and automatic bad-frame rejection, turning days of manual collection into a single unattended run.
- Brought up mainline Linux 7.1 and Arch Linux ARM on the PolyOculus telescope array's eight Jetson Nano nodes, replacing NVIDIA's end-of-life Ubuntu 18.04 image: traced boot failures to an SD-card regulator behind a modular Tegra I2C driver, U-Boot's 14 MiB initramfs load window, and missing Tegra PWM/host1x drivers, and built a kernel that boots with no initramfs. Any node rebuilds in about 15 minutes (Ansible).
- Wrote the node control software: centroid-based autoguiding from a USB3 guide camera (driven through its vendor UVC extension unit for 1 µs exposure control) to INDI mount guide pulses, cutting simulated tracking error from 3.01″ to 0.29″ RMS, plus a closed-loop piezo steering mirror that holds a laser spot to 0.2 px.

SELECTED PROJECTS

CELERIS — GPU-accelerated numerical solver. C++23, CUDA, Python 2026

- Wrote a Maxwell solver and design pipeline from scratch, including a CUDA kernel benchmarked at 4.8–5.8× over the project's own optimized 16-core CPU path at 92k elements and diagnosed as memory-bound from the way the speedup declined with problem size. Gradient-based (Adam) optimization over high-dimensional design parameters; Monte-Carlo tolerance analysis. *Sole-author manuscript in preparation.*

Stardust — finding new near-Earth asteroids in archival sky-survey images. PyTorch, CUDA, Python October 2026

- Knight Hacks IX team project; I built the detection and orbit pipeline. A batched GPU N-body integrator I wrote in PyTorch samples a 4-day orbit's uncertainty (100,000 clones × 1 year in 23 s on a laptop RTX 4070) and projects it onto years of pre-discovery ZTF images; detections must link into one track that survives an orbit refit.
- Trained a 1.9M-parameter U-Net detector on 4,000 real ZTF cutouts, after catching a 3–4″ error in an upstream service's labels by checking them against the pixels. On unseen nights at a matched false-alarm rate it finds 57.5% of asteroids at the camera's limit, where a classic streak filter finds none. Blind backtest on 72 asteroids NASA once listed as possible impactors: 167 verified, never-connected sightings of 22, tightening their orbits up to 52× (median 7.1×); a frozen held-out round recovered 16 of 50.

TECHNICAL SKILLS

Machine learning: empirical evaluation and benchmarking methodology, adversarial robustness, LLM inference and prompting at scale (Azure OpenAI batch API), agentic pipelines, transformer-based classification, perplexity and coherence metrics, reinforcement learning, tree-search decoding

Numerical and systems: CUDA kernel development and profiling, embedded Linux bring-up on NVIDIA Jetson (kernel config, U-Boot), closed-loop control, HPC cluster work, linear algebra, FFT and signal processing, gradient-based optimization, Monte-Carlo methods

Languages and tools: Python (NumPy, SciPy), C++ (C++23), CUDA, Rust, C, Go; Linux, Git, CMake, Docker, LaTeX/Typst

HONORS

Florida Bright Futures — Florida Academic Scholars (highest tier; 100% tuition) 2025